Astro Night Planner 1.4.0-test.1

Practical guide for planning astrophotography nights with weather, twilight, Moon, object selection, equipment, horizon and framing

Updated: 22 June 2026

Important note on settings, cookies and backup

The Astro Night Planner stores profiles, equipment, locations, horizon profiles and settings locally in IndexedDB. Classic cookies are not the primary storage location.

The installable web app files are stored separately in Cache Storage. Deleting only the browser cache usually does not delete IndexedDB data; deleting site data can remove settings, profiles and stored file permissions.

Persistent storage protects only against automatic browser cleanup, not against deliberate deletion by the user. Regular external backups remain necessary.

Contents

What’s new

Version 1.4.0-test.1

- Aurora assessment was corrected: NOAA alert messages no longer set a local warning color by themselves.
- NOAA Kp, GFZ Kp forecast and NOAA messages are shown separately in the dashboard.
- The aurora dashboard now includes refresh, NOAA/GFZ reference graphics, enlarged view and source links.
- The help entry “What’s new” now shows the current version first and older version steps below.

Overview and concept

The Astro Night Planner is a practical decision tool for deep-sky astrophotography. It does not only answer whether an object is theoretically visible; it combines visibility, twilight, Moon, weather, personal equipment, location, horizon and framing into one object rating.

Planning

Location, date, temporary profiles, weather, cloud map, object filters, object list, altitude curve, horizon view and Aladin framing.

Settings

Equipment, locations, horizon profiles, display, rating logic, weather profiles, backup, help and legal information.

Location → **Date** → **Weather** → **Filters** → **Object details** → **Framing**

> **Benefit:** You no longer need to check a weather app, star atlas, framing tool and object list separately. The Planner brings these steps into one planning flow.

Version overview

This overview summarizes the changes in version 1.2.0 compared with 1.1.0 and also the main changes in version 1.1.0 compared with 1.0.0.

Version 1.2.0 compared with 1.1.0

- The separate weather comparison in a new tab has been removed.
- Windy, Ventusky, Meteoblue and Clear Outside remain available as individual external reference sources under “Additional weather sources”.
- Each external weather source uses its own controls and timeline; the app no longer displays a shared comparison time for those external maps.
- The own astro cloud map remains available only in the planning view and is no longer rebuilt in a separate comparison window.
- No longer needed code for the comparison window, shared comparison time and time handoff to external maps has been removed.
- Help, handbook and version notes have been adjusted to the more stable handling of weather sources.

Version 1.1.0 compared with 1.0.0

- PWA with local user profiles, separate test/production storage and backup/restore functions.
- Extended equipment model: cameras with sensor and resolution data, telescopes and lenses with focal length and focal ratio, plus reducer, flattener and Barlow factors.
- Improved Aladin framing with movable camera frame, information boxes, Moon display, object labels, external sky-image view and N.I.N.A. export.
- Locations and horizon profiles have been extended, including interactive horizon editor and N.I.N.A. import/export.
- Object selection extended with direct search without other filters, configurable object-list information, personal imaging targets and additional filter options.
- Weather section extended with astro cloud model, hourly weather trend, model consensus, cloud layers, precipitation/rain/snow and additional external weather sources.
- Help, browser data FAQ, update handling and tablet/iPad operation have been improved.

First steps: recommended setup

1. Open **Settings - Equipment** and check telescope, camera and mount.
2. Create your imaging location under **Locations & horizon**. Use search or GPS, but verify the coordinates.
3. Enter your personal horizon. Even a rough profile is better than assuming a completely flat horizon if trees, buildings or hills interfere.
4. Check defaults for weather rating, display and cloud map under **Central settings**.
5. Switch to Planning, select date and location, and check weather and cloud map first.

6. Filter the object list by catalogs, types, minimum altitude, visible duration, Moon distance and size.
7. Open an object and assess altitude curve, horizon view and framing.
8. Set up an external backup before creating many profiles and locations.

Example: For a short evening session choose the nautical window, set visible duration to 2 hours, minimum altitude to 25 degrees and Moon distance to 40 degrees. Then inspect the best objects by framing and cloud evolution.

Language and documentation

On first launch the app checks the browser or device language. German starts in German; all other languages start in English. After manual switching, the saved selection is used.

The title bar contains the switch 🌐 **DE | EN**. It changes the active language and stores the choice locally. Help and PDF links automatically open the corresponding language version.

> **Note:** The German legal texts are authoritative. English texts are provided as an additional aid.

Local data, cache and backup

Profiles, equipment, locations, horizon profiles and settings are stored in **IndexedDB**. Installable app files are stored separately in **Cache Storage**. Classic cookies are not the primary storage location.

Browser action	Typical effect
Delete only cached images and files	Usually removes PWA files, but not IndexedDB data.
Delete site data	Can remove profiles, settings, file permissions and cache.
Request persistent storage	Protects against automatic cleanup, not deliberate deletion.

> **Recommendation:** Use *Export all data* regularly or enable automatic backup. `astro-night-planner-aktuell.json` is the current state; dated files are historical restore points.

Location, date and planning window

Location and date in Planning apply to the current planning session. The default location is set in Settings. A planning night starts on the evening of the selected date and ends the next morning.

Window	When useful?
Sunset to sunrise	Overview curves and broad object paths.
Nautical planning window	Recommended default for many deep-sky plans because bright twilight edges are reduced.
Astronomical night	For demanding faint targets and very dark conditions.

The displayed curve can be wider than the rating basis. The current label **minimum visibility** technically means a minimum duration above the minimum altitude and personal horizon. It will later be renamed more clearly.

Profiles for this plan

This section allows temporary deviations without overwriting saved defaults. This is useful when testing another telescope, camera, horizon profile or weather profile for one night.

- **Telescope/camera:** affect field of view, framing and framing rating.
- **Horizon profile:** affects visible duration and horizon view.
- **Imaging quality profile:** affects wind and gust evaluation.
- **Display profile:** controls object-list columns, order and page size.

Practice: On a windy night you can temporarily test a more robust wind profile without changing the normal default.

Weather and imaging quality

The app combines weather models into a consensus. Large KPI fields summarize the night; the hourly table shows when conditions change.

Criterion	Meaning for imaging
Clouds	Main exclusion factor. High clouds can strongly degrade transparency.
Effective transparency	Combines cloud cover and atmospheric clarity.
Seeing	Tendency from jet stream and surface wind; not an exact arcsecond measurement.
Wind/gusts	Expected imaging quality, not mechanical safety.
Dew gap	A small gap increases dew risk.
Moon	Influence via distance, illumination and altitude.

> **Interpretation:** A good total score does not replace checking individual values. A 75/100 night may be good for narrowband but poor for faint galaxies due to Moon or transparency.

Astro cloud model

The astro cloud model is its own planning section. It can be opened or collapsed like the hourly weather trend. The section shows cloud distribution, model disagreement and optional precipitation layers around the selected planning location.

- Percent values are shown at the actual forecast points.
- Smoothing changes only the display, not the underlying weather data.
- 15-, 30- or 60-minute steps interpolate between hourly model values and do not increase model accuracy.
- **Total precipitation** includes rain, showers and snow. **Rain** and **Snow** show the individual components.

Cloud-edge example: The value at your location may look good while a denser cloud edge lies to the west or south-west. If the estimated movement points toward your location, a long imaging run is riskier than the single number suggests.

Location-comparison example: When conditions are uncertain, compare saved locations. If Filderstadt shows 60% cloud cover but a mobile site in the Black Forest shows 20%, inspect the map distribution and model disagreement before deciding.

Additional weather sources

Meteoblue, Clear Outside, Windy and Ventusky are used as independent reference sources and are not included automatically in the model consensus. They are useful when the Open-Meteo models disagree or when you want to check cloud, precipitation and wind trends separately.

The sources are embedded on the planning page as separate tabs. Each source is loaded only when its tab is opened. The embedded Meteoblue view starts as a quick wind-animation map; clouds and precipitation are opened deliberately through the separate Meteoblue buttons. Windy and Ventusky open interactive maps centered on the planning location, while Clear Outside provides a compact forecast image.

Practical example: Before a long drive, first check the astro cloud model. If the situation is borderline, open Meteoblue, Windy, Ventusky or Clear Outside one after another and manually compare whether cloud bands, precipitation and wind direction match your own map.

> **Note:** The former separate weather comparison view with a shared time control has been removed. Embedded external maps do not reliably accept one central time, so each source keeps its own timeline and controls.

Combining object filters

Filters should reduce the object list quickly to useful targets. Start broad and tighten only afterwards.

Reset basic filters restores only the values in the Basic filters section to their defaults. Catalogues, imaging filters, object types, equipment, location, display profile and other planning settings remain unchanged. This is useful when temporary filters for one night have made the object list too restrictive.

Filter	Recommendation
Catalogs	Activate only catalogs you really want to see. Special catalogs can create very large result sets.
Object types	Narrowband: emission nebulae, planetary nebulae, supernova remnants. Broadband: galaxies, clusters, reflection nebulae.
Minimum altitude	25 degrees is a good starting point. Lower can work with a free horizon; use more with light domes or obstacles.
Minimum visibility	Prevents targets that are usable only briefly. It means minimum duration above horizon/minimum altitude.
Moon distance	Be strict for faint broadband targets, more relaxed for narrowband.
Object size	Helps find targets that match your field of view.

Narrowband practice: Catalogs NGC/IC/Sh2, types emission nebulae and supernova remnants, filters Ha/OIII/SII, moderate Moon distance, then check framing.

Object list, score and mini altitude profiles

The object list is sorted by score. The score combines weather, altitude, Moon, visible duration and other criteria according to the active profile. Mini altitude profiles show immediately whether an object is useful for a longer period or only briefly near the edge.

Columns can be shown, hidden and sorted per display profile in Settings. Compact profiles are useful on mobile; the detailed view can show more information on desktop.

> **Do not blindly trust rank 1:** Always open details for your favorites. Framing, object size and cloud evolution can change the practical decision.

Object details, altitude curve and horizon

Clicking an object row opens details below. Altitude curve and horizon view use the same time control, so altitude, azimuth, twilight, horizon clearance and weather values can be assessed together.

The horizon view is essential if your local horizon is not flat. An object may be astronomically 30 degrees high but still behind a tree or building.

Example: A target with a high score shortly after twilight may be unusable if it clears your local obstruction only above 35 degrees in that direction.

Framing with Aladin Lite

The Aladin view shows the sky image, camera frame and - where available - a curated object outline. If no outline is available, the app falls back to the technical ellipse based on catalogue size and position angle. Camera frame, comparison frames, object outline, Moon marker and labels are overlay layers above the native Aladin survey.

The Moon is displayed as a pixel-stable symbol above Aladin. Position, label and phase use the same calculated Moon RA/Dec coordinate. The illuminated part is white to warm white, while the unlit side remains transparent or very dark-transparent. Moving the **Time in planning window** slider updates position, illumination and phase.

- **Frame to image center:** places the camera frame on the currently visible center after panning.
- **Reset object orientation:** restores the technical ellipse to the catalogue position angle if an object has no curated outline.
- **Time in planning window:** does not change the star field; it updates time-dependent values such as altitude, direction, twilight, weather and the optional Moon position.
- **Show object names:** displays labels depending on zoom level.
- **Open sky image in new tab:** opens the Aladin detail map larger in a separate browser tab.

> **Why the star field does not rotate:** In an equatorial framing context, object coordinates and camera angle remain fixed relative to the sky. Time mainly changes altitude, azimuth and observing conditions.

Aladin surveys, setups and measuring tool

Use the native survey rendering

Aladin surveys are loaded in their native rendering. A survey such as **DSS2 red** must therefore not be artificially colored red: the name describes the red photographic band, while Aladin's native visual presentation is black and white. The Planner does not apply an artificial colormap to monochrome surveys. Surveys appear in color only when the source itself provides a color survey or color composite.

Example survey choice: Galaxies usually work well with an optical color survey or DSS2 red. Emission nebulae can be easier to judge in H-alpha. Dust and dark nebulae may benefit from infrared or dust surveys. When adding a survey, always check whether the wavelength range really answers your planning question.

Add custom surveys

Under **Settings - Central settings - Display** you can choose which Aladin surveys appear in the object map. You can also add custom HiPS surveys manually. A display name and the technical HiPS ID or a suitable HiPS URL are required. Custom surveys remain stored locally in your profile.

Field	Recommendation
Display name	Use a short, clear name, for example *NSNS H-alpha* or *My Ha survey*.
HiPS ID	Use the exact survey ID or URL. Invalid IDs prevent Aladin from loading an image.
In selection	Only enabled surveys appear in the compact object-map selection.

Setup combinations and comparison frames

Planning uses one active setup consisting of telescope and camera. This setup controls field of view, pixel scale, framing rating and filters such as *fits on sensor*. In the Aladin detail view you can additionally show up to three setup combinations as local comparison frames. These frames do not change the global object list and only affect the currently opened sky image.

Configuration example: Create *Askar 200 + QHY268M* as your standard setup. Add a second setup, for example *Lacerta + QHY268M*. Planning still uses one active setup; in Aladin you enable the second setup only as a comparison frame to see immediately whether the object fits better at another focal length.

Measuring tool in the external tab

The external Aladin tab includes a measuring tool. After two clicks it shows the angular distance numerically and as a visible line with endpoints in the sky image. This is useful for object separations, rough mosaic planning, Moon distance checks and comparing neighbouring nebula structures.

Practice tip: Use the embedded map for quick framing and the external tab when you want more space for measuring, checking surveys or inspecting details.

Local surveys and Local Survey Server

The app can still load Aladin surveys directly online by HiPS ID or HiPS URL. For offline planning you can additionally configure a local HiPS source. A normal Windows or macOS file path cannot be read directly by the PWA; the local survey data must be served through a local HTTP address.

Example: Create a root folder `D:\AstroSurveys`. Several HiPS surveys are stored below it, for example `nsns/ohs8/`, `nsns/halpha8/` and `dss2/red/`. A local survey server exposes the root folder as `http://127.0.0.1:8765/`. Enter that address as the local survey base URL in the app; per survey only the relative path such as `nsns/ohs8/` is stored.

1. Create a local survey root folder and place HiPS data there.
2. Start a local web server or later the optional *Astro Night Planner Local Survey Server*.
3. Enter the local base URL under **Settings - Central settings - Display - Aladin - Survey - Local survey sources**.
4. Use **Local source** for each survey to check or adjust the relative path.
5. Use **Check local survey source** to test whether the source is reachable. This button does not start the server.

If local sources are enabled and a survey source is reachable, the app can prefer that local source. If it is unavailable, the online source can remain available as fallback. A single local web server is sufficient for all surveys if all survey folders are stored below one common root folder.

> **Important:** The PWA cannot start a local web server for security reasons. Comfortable offline use is planned through a separate helper program that starts the local survey server by double click or autostart.

Equipment

Telescopes, cameras and mounts are stored locally. Telescope and camera determine field of view, image scale and framing rating. The mount is currently documented and may later be used for more detailed quality profiles.

Field	Why important?
Focal length	Determines field of view together with sensor size.
Sensor width/height	Determines whether an object fits into the camera frame.
Pixel size	Basis for sampling and future detail ratings.

Locations and horizon profiles

Locations contain coordinates, elevation, time zone and horizon profiles. Search results should be checked deliberately, especially for postal codes or places with identical names. The time zone is stored as an IANA time zone.

Several horizon profiles per location can be useful: garden, terrace, balcony, mobile field or observatory. A profile can be selected temporarily in Planning without changing the default.

Practice: A location can be free to the south but blocked by trees to the north-east. This strongly affects M31, M45 and early autumn targets.

Central settings and defaults

Central settings define defaults, not necessarily the current plan. Many values can be temporarily overridden in Planning.

- **Quality traffic light:** default red 0-59, yellow 60-79, green 80-100.
- **Deep-sky weighting:** default: clouds 30%, transparency 15%, seeing 10%, wind 10%, dew 10%, Moon 10%, altitude 10%, duration 5%.
- **Wind profiles:** describe expected imaging quality for different setups.
- **Cloud map:** default mode, smoothing, grid and percent values.
- **Display profiles:** object-list columns, order and page size.

Saving: Many settings sections apply changes only after saving. This protects against accidental permanent changes.

New filters, target list and settings tabs

The object filters have been extended so the list can better match the actual imaging situation. Many values are split into persistent defaults in Settings and temporary choices in Planning. This lets you adapt a single night without changing your normal setup.

Minimum score

The **Minimum score** field hides objects below a given quality value. The default is **0**, meaning no restriction. A value of 60 keeps only targets that reach at least the yellow quality range.

Minimum visible duration

This is the minimum time during which an object is above the minimum altitude and your personal horizon. Its reference period can be selected separately. The default is the nautical planning window so bright edge periods are not accidentally counted.

Object size in degrees and arcminutes

Minimum and maximum object size are entered with separate degree and arcminute fields. The app still calculates internally in arcminutes. Minute values above 59 are prevented or normalised. Size presets can be stored in Settings and adjusted temporarily in Planning.

Example: The default range 0°03′ to 4°00′ removes very tiny specialist objects while keeping many classic deep-sky targets for medium focal lengths.

Surface brightness and missing values

The surface-brightness filter helps identify very faint extended targets. Because not every catalogue object has reliable data, objects without a surface-brightness value remain visible by default. Strict filtering can hide them deliberately.

Imaging filters and object-type profiles

The recommended filters L, R, G, B, Ha, OIII and SII can be used as a multi-select filter. The additional option **no filter information** prevents incomplete catalogue entries from disappearing unnoticed. Object-type combinations can be saved as profiles, for example galaxies, nebulae or narrowband targets. Manual changes in Planning remain temporary.

Framing filter

The framing filter distinguishes well-framed, near-edge, too-large and unknown objects. It uses the active telescope/camera setup and is helpful when you want targets that fit without a mosaic.

New catalogues and My imaging targets

Barnard and LDN/LBN can be enabled as additional catalogues; they are initially inactive. The personal catalogue **My imaging targets** is also initially inactive. You can mark objects, add notes, priority, status and reference links. When this is the only active catalogue, Planning answers the practical question: *What can I image tonight that I have always wanted to capture?*

Moon in the object map

The Moon can optionally be displayed in the Aladin view. Its position is calculated for the selected time in the planning window and updates when the time slider moves. It is initially disabled to keep the framing view clean.

Settings tabs

Settings are split into tabs. Central settings, locations/horizon, info/help/backup and My imaging targets are easier to navigate. The object-list configurations Compact, Standard and Detailed remain visible and no longer collapse automatically when columns are moved.

Export, import and restore

Export all data includes all profiles, equipment, locations and settings. **Export current profile** transfers a single profile. **Import profile** creates a new imported profile and does not overwrite existing profiles without confirmation.

1. Create a full backup regularly.
2. Store the file outside the browser profile.
3. Use `astro-night-planner-aktuell.json` for normal recovery.
4. Use dated files only when deliberately returning to an older state.
5. After restore, check active profile, location and equipment.

PWA installation and updates

The app can be installed as a PWA. App files are available locally; weather, maps, catalog updates and external sky images still require an internet connection.

When a new version is available, an update banner may appear. After updating, fully closing and reopening the installed PWA can be necessary. Deleting site data is not needed for normal updates.

New in Test 18: surveys, locations, object details and outlines

NSNS surveys for narrowband planning

The Northern Sky Narrowband Survey (NSNS) is available as a survey group. H-alpha, OIII, SII and OIII/H-alpha/SII are enabled in the Aladin selector by default. H-alpha is useful for emission nebulae, OIII for planetary nebulae and supernova remnants, and SII for SHO planning. Additional NSNS variants can be enabled in Settings.

Adding locations clearly

Use **Add location** to open a dedicated input area for a new imaging site. After search or manual coordinate entry, **Save location** stores the site, selects it immediately and makes it available in the planning view. Existing location changes are stored with **Save location changes**.

Clickable objects in the Aladin view

Objects from the Planner catalog overlay can be clicked in the Aladin image. The popover shows catalog number, name, type, coordinates, size, magnitude and catalog membership. From there you can frame the selected object or add it directly to **My imaging targets**.

Manual object outlines

Open sky view in new tab opens the larger Aladin view with measuring tool and outline drawing. The drawing mode can be **Control points** or **Freehand**. Control points are the recommended default because they are easier to correct. Freehand drawing is smoothed and internally reduced to coordinate points. Saved outlines are stored locally in IndexedDB, loaded automatically when the outline option is enabled, and included in full backups. If no outline is available, the app falls back to the catalog ellipse.

Wikipedia

The **Wikipedia** button opens a Wikipedia search in the active app language. The search now prefers catalog identifiers such as **SH 2-119**, **NGC 7000** or **IC 5070**. Common names and aliases are used only as fallback terms. The app does not force a potentially wrong article; unclear targets open search results instead.

Legal notice, privacy and usage notes

The footer provides access to legal notice, privacy, usage notes, data sources & licenses, help and version. The app does not integrate its own analytics, tracking or error reporting. GitHub Pages and external services can process technically required connection data.

Provider:
Andreas Cordt
Schwarzwaldstraße 15
70794 Filderstadt, Germany
Email: andreas@deepskyastrophoto.de
Homepage: www.deepskyastrophoto.de

Weather and calculation values are planning aids. Safety decisions must be based on local conditions and official warnings.

Troubleshooting

Problem	Recommendation
Old display or old help	Reload the app, accept the update prompt, fully close and reopen the installed PWA.
Weather data missing	Check internet connection and try later; external services can be temporarily unavailable.
Map does not appear	Repeat date change, collapse/open the map section or reload the page.
Aladin does not load	Reload sky image, check network, try the external service later.
Profiles are missing	Check whether site data was deleted; restore the latest external backup.

Glossary

Term	Meaning
IndexedDB	Browser database for profiles and settings.
Cache Storage	Storage for PWA app files.
Nautical window	Period with the Sun below about -6 degrees.
Seeing	Atmospheric turbulence that broadens or distorts stars.
Atmospheric transparency	Cloud-independent air clarity. Aerosols, haze, moisture, thin veil and extinction can reduce faint structures even under apparently cloud-free skies.
Effective transparency	Practical main value for astrophotography that combines atmospheric clarity and cloud cover.
Framing	Fit of object size and camera field.
Minimum visibility	Current label for minimum duration above minimum altitude and horizon; it will be renamed later.

Version 1.0.2 additions

This production version corrects the 1.0.1 production package and still includes the 1.0.1 changes.

- The main navigation buttons **Planning** and **Settings** are now visibly present in the production package on the right side of the fixed title bar. The former bottom button bar is removed, leaving more vertical space for tables, hourly weather and detail views.
- The LDN catalog has been extended with 1787 named Lynds dark nebulae. Search now finds entries such as `LDN 1093` and `LDN1093`.
- The existing **LDN/LBN** catalog filter selects the LDN entries in this version. LBN is not yet imported as a separate catalog.
- LDN sizes are derived from the catalogued area as an equivalent circular diameter. This is a practical filtering and framing aid, not an exact visible object boundary.
- The local catalog cache key was increased so browser profiles load the new catalog without manually clearing site data.

Version 1.0.0 development additions 23

This correction further improves the Moon display in the Aladin object map and adds a targeted reset for the basic filters.

- The Moon symbol is no longer drawn as a sky-coordinate circle, but as a pixel-stable SVG/DOM symbol. Its size therefore remains constant while zooming.
- Symbol, label and phase are anchored to exactly the same calculated Moon position. This avoids the previous offset between marker and symbol.
- The illuminated Moon area is white to slightly warm white; the unlit side is transparent or very dark-transparent.
- The **Time in planning window** slider still updates Moon position, illumination and the information panel.
- Next to **Apply filters** there is a new **Reset basic filters** button. It resets only the basic filters and leaves catalogues, imaging filters, object types, equipment, location and display profiles unchanged.

Version 1.0.0 development additions 22

This correction improves the Moon display in the Aladin object view and fixes two layout details in the basic filter.

- The Moon is now shown as a small drawn Moon symbol with illuminated portion instead of a target circle or crosshair.
- The label next to the symbol always reads **Moon**. In the German interface it reads **Mond**.
- The symbol uses the illumination calculated for the planning time and approximates crescent, quarter, waxing or waning gibbous and full Moon phases.
- Moving the **Time in planning window** slider updates the Moon position and phase symbol for the selected time.
- The note for the **visible-duration reference** is kept on one line on wide screens.

- Number fields no longer show up/down spinner arrows in Firefox; Chromium/WebKit spinners remain hidden as well.

Version 1.0.0 development additions 21

This correction adds the missing direct object search, makes numeric filter fields more compact and improves Wikipedia search terms.

- The basic filter now has two search fields: **Search within active filters** and **Direct search without other filters**.
- The normal search refines the currently active filters. Direct search ignores catalogs, object types, size, altitude, minimum score, Moon distance and framing filters. It is intended for quickly finding a known object even when other filters would currently hide it.
- Examples for direct search: `SH 2-119`, `Sh2-119`, `NGC 7000`, `IC 5070` or an alias.
- Active search fields are highlighted. When direct search is active, the app also shows a note that other filters are ignored.
- Numeric fields in the basic and size filters are more compact so the rows are calmer and the size fields line up more cleanly.
- Wikipedia searches now prefer normalized catalog identifiers such as `SH 2-119`, `NGC 7000` or `IC 5070`. Common names are used only as fallback terms when no useful identifier is available.

Atmospheric and effective transparency

Atmospheric transparency describes cloud-independent air clarity. Haze, humidity, aerosols, Saharan dust, smoke or thin veil can lower the contrast of faint nebula and galaxy structures even when the cloud map looks good.

Effective transparency is usually the more practical value for an imaging decision. It combines atmospheric clarity with actual cloud cover. Excellent air clarity is of little use if clouds pass through; conversely, under cloud-free skies, mediocre atmospheric transparency can explain why faint structures look weak or washed out.

Version 1.0.0 development additions 20

This correction improves collapsible controls, the object list and the manual object-outline planning flow.

- Collapsible areas such as the Aladin survey list and the object-list information settings now show a clear disclosure arrow.
- The basic filters use aligned field rows so inputs line up more cleanly.
- Wikipedia is available as a configurable object-list information item. Clicking the Wikipedia icon opens Wikipedia; clicking other parts of the object row still opens the details.
- Catalog numbers and aliases in the Aladin object popup are de-duplicated so identifiers such as SH2-119 do not appear twice and Wikipedia search terms remain clean.
- The Aladin object popup now has a close button.

- Custom object outlines are saved and loaded more robustly. Saved outlines are shown in orange/yellow without visible control points; the circle or ellipse is used only as a fallback when no saved outline exists.
- The external Aladin tab can delete a saved custom object outline.

Addendum 1.1.0

- The update button now activates new app files more reliably, clears older app caches and reloads with cache busting.
- The horizon editor has been revised again and uses direct mouse, pen and touch input.
- N.I.N.A. horizon import opens a file dialog for TXT/HRZ/CSV horizon files; export downloads an HRZ text file.
- The first-run screen now includes a button to restore a full backup directly.
- The help now contains a browser data FAQ explaining Firefox, Chrome/Edge, Safari/iOS and settings that can delete local data on browser close.

Addendum 1.1.0

This version extends the settings, equipment model and Aladin framing tools.

- Tablet and iPad layouts were made more robust for wrapped header, form, filter and Aladin control areas.
- New-entry placeholder names such as "New location", "New camera" or "My telescope" are easier to replace on touch devices: the default text is selected on first focus.
- Cameras can now store sensor size, resolution and megapixels. If sensor size and resolution are available, the app can derive the pixel size.
- Telescopes and camera lenses can be maintained by focal length and f-number. The aperture is calculated from both values.
- Reducer, flattener and Barlow factors can be assigned to setups. The effective focal length is used for framing, field of view and N.I.N.A. export.
- Long settings pages now provide an additional save action at the top.
- Moonrise and moonset hints include the actual time when the event lies outside the selected planning window.
- In the Aladin framing view, info boxes can be shown or hidden and positioned left, right, top or bottom.
- The camera frame can be moved independently of the object center; the N.I.N.A. export uses the current frame center.
- The app warns about unsaved settings changes when the user leaves the settings page, changes settings tabs or closes/reloads the browser tab.

Version 1.2.0

This corrective version removes the separate weather comparison view completely.

- The **Open weather comparison in new tab** button has been removed from the **Additional weather sources** section.
- The own astro cloud map remains available in the planning view; it is no longer rebuilt in a separate comparison window.
- Windy, Ventusky, Meteoblue and Clear Outside remain available as individual tabs and each keeps its own controls and timeline.
- The no longer needed comparison-window code, shared comparison time and related synchronization messages were removed from the application.
- Help and handbook now describe the external weather sources as individual reference sources instead of a 2x2 comparison.

Addition 1.4.0-test.1

This test version refines the experimental sky and horizon functions in the external Aladin tab. Framing remains the primary purpose; additional ground and grid elements are orientation aids.

- **Ground transparency:** The setting is now interpreted as true transparency. 0% means strongly darkened, 50% means semi-transparent, and 100% means fully transparent.
- **Azimuthal grid:** In Auto mode, the app adapts the spacing to the zoom level. Wide views use coarser lines; stronger zoom uses finer spacing down to 1 degree. The spacing can also be fixed to 10°, 5°, 2° or 1°.
- **Degree labels:** The azimuthal grid shows degree labels at visible line ends or near the view edge, making individual lines easier to interpret.
- **Compass directions:** N, E, S and W plus intercardinal directions such as NE, SE, SW and NW are displayed more robustly. If the true direction lies outside the field of view, the label is clamped to the visible edge.
- **Color and line width:** The azimuthal app grid remains configurable by color picker and line width. The color swatch shows the selected color more clearly than the native browser picker alone.

Update 1.4.0-test.1

This test version corrects the orientation aids in the external Aladin tab.

- **Azimuthal grid:** Degree labels are placed at the visible edge of grid lines even at higher zoom levels and fine spacing.
- **Compass directions:** N, E, S and W plus NE, SE, SW and NW are displayed more reliably at the visible edge.
- **Color selection:** The azimuthal grid now uses a single color field; the selected color is visible directly in that field.

- **Horizon display:** The standard horizon may appear as a straight line in the Aladin projection. The ground/horizon display is an orientation aid in the external tab; framing remains the primary function.

Version 1.4.0-test.1 additions

This test version refines the new 1.3 features. Aurora assessment is not fixed to Filderstadt or Germany; it relates to the currently selected location. The configured Kp thresholds are adjusted in a simplified way by geographic latitude: higher latitudes can require lower activity, lower latitudes require stronger events.

- **Aurora indicator:** Yellow means increased geomagnetic activity, orange means a plausible chance at the selected location, red means a strong situation. A mere count of NOAA messages without an extractable G or Kp strength does not create a local warning color.
- **Aurora dashboard:** shows data status, source, refresh time, reasoning, Kp forecast and detected messages separately.
- **Object list:** horizontal scrolling is retained even when object details are open, so buttons on the right remain reachable.
- **Aladin - Sky & Horizon:** the native Aladin grid remains controlled inside Aladin. In addition, the external Aladin tab can show an app-generated azimuthal grid with configurable color and line width.
- **Ground display:** the external Aladin tab offers no ground, standard horizon (0 degrees), or personal horizon. If no personal horizon profile is available, the standard horizon is used as fallback. Ground opacity is configurable.
- **Altitude information:** the Aladin info box separates altitude above mathematical horizon and above personal horizon.

Version 1.4.0-test.1 additions

- The wide object-list table and the object details are separated so the horizontal scrollbar stays at the parent object list.
- Long visibility column headings wrap to several lines.
- Aurora warning colors are easier to distinguish by color, background and state label.
- The external Aladin tab draws ground/horizon and the azimuthal grid as a visible overlay above Aladin and recalculates it during pan and zoom.
- The azimuthal grid color setting now shows an additional color swatch.

Version 1.0.0 development addendum 19

This correction version improves location management, Aladin object interaction and manual outline drawing.

- When adding a new location, latitude and longitude initially stay empty. Search and GPS are available only inside “Add new location”.
- The basic-filter search field is now placed in the last filter row close to the object list.
- The Aladin object popover shows more object information and the frame can be moved to the selected object without intentionally closing the popover.
- The “Open details in planning” action has been removed from the Aladin popover.
- During outline drawing, control points and connecting lines are drawn visibly. After saving, the custom outline is preferred over the ellipse fallback.
- Survey management is an initially collapsed expert list at the bottom of the display settings section.

Additions in version 1.1.0

Calendar date and long-term planning

A calendar date can be selected in addition to the quick forecast-day buttons. If the selected date is outside the available forecast range, weather-dependent sections are hidden and a notice is shown. Astronomical object planning, Moon information, altitude trends and framing remain available.

Show or hide sections

The display settings can now define which planning sections are shown at all. This is independent of the default expanded/collapsed state.

Horizon and N.I.N.A.

The horizon editor shows azimuth and altitude while drawing. N.I.N.A. horizon files with azimuth/altitude pairs can be imported, and current horizon profiles can be exported.

N.I.N.A. export from framing

The currently opened object in Aladin can be exported together with the current frame center, rotation, FOV and setup data as a JSON file. The export intentionally refers to the single opened object only.

Catalogs

LDN and LBN are offered as separate catalog filters. LDN data is included. LBN and Barnard are prepared as separate catalog areas; the complete datasets will be added after a separate raw-data verification.